PART 4: SEQUENTIAL LOGIC
ASYNCHRONOUS COUNTERS

SAAD BAIG – DIGITAL LOGIC DESIGN

The animations and contents in these slides are originally adapted from the DLD course slides designed by Dr. Hasan Baig and Mr. Junaid Ahmed Memon. The original slides can be accessed from https://www.hasanbaig.com/resources

© Hasan Baig and Junaid Ahmed Memon. All rights reserved. These files and accompanying materials are made available under the terms of "Attribution-NonCommercial-NoDerivatives 4.0 International (CC BY-NC-ND 4.0) License" and is available at https://creativecommons.org/licenses/by-nc-nd/4.0/

COUNTERS

- A register that goes through a prescribed sequence of states upon the application of input pulses at random or fixed interval of time.
- Input pulses are clock pulses that originate from some external source.
- The sequence of states may follow the binary number sequence (making it a binary counter) or any other sequence of states.

COUNTER CATEGORIES

- Ripple counters:
  - The flip-flop output transition serves as a source for triggering other flip-flops.
  - No common clock pulse (not synchronous).

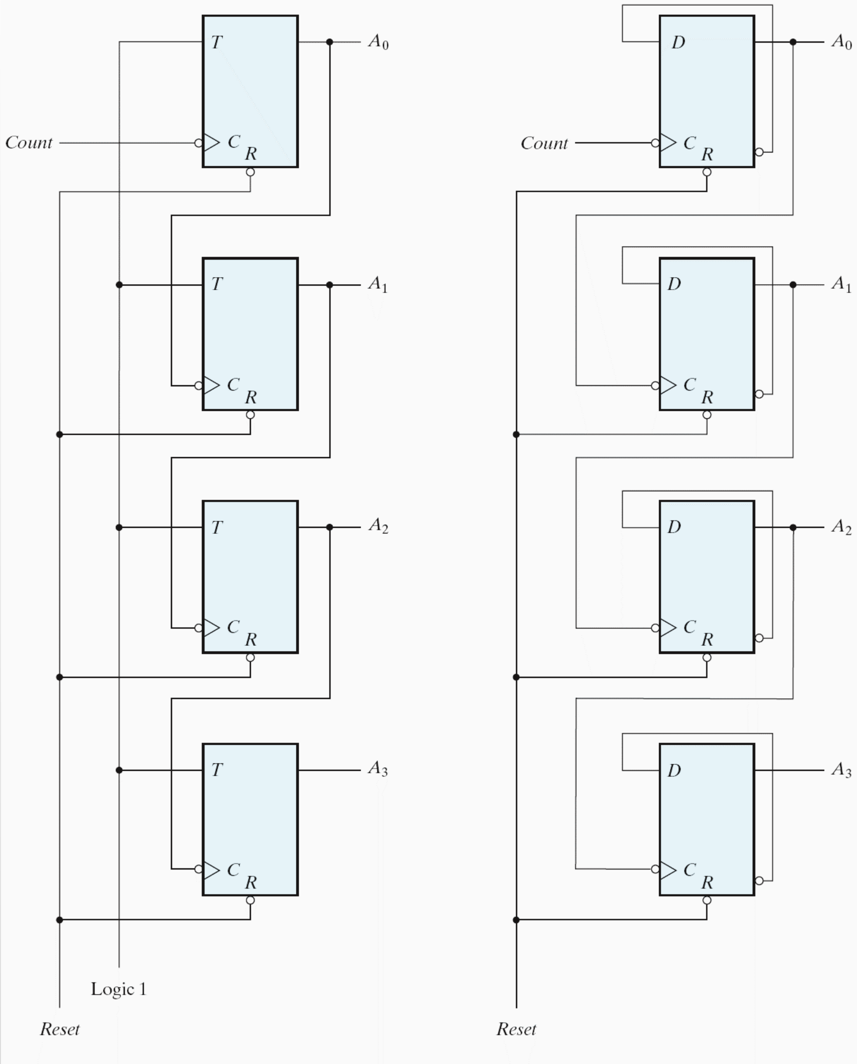
- Synchronous counters:
  - The CLK inputs of all flip-flops receive a common clock

Binary Count Sequence

A_3	A_2	A_1	A_0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0

- It consists of a series connection of complementing flip-flops. The output of each flip-flop is connected to the *C* input of the next higher order flip-flops. It will count from 0 to 15 and then back to 0.

4-BIT BINARY RIPPLE COUNTER



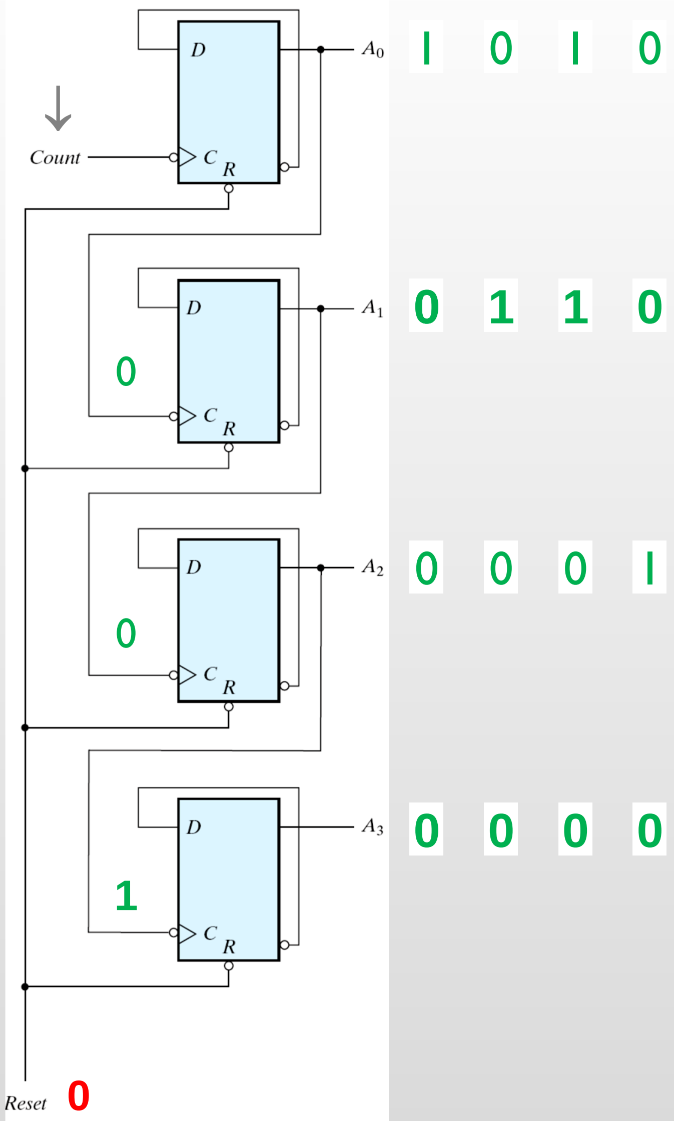
(a) With *T* flip-flops (b) With *D* flip-flops

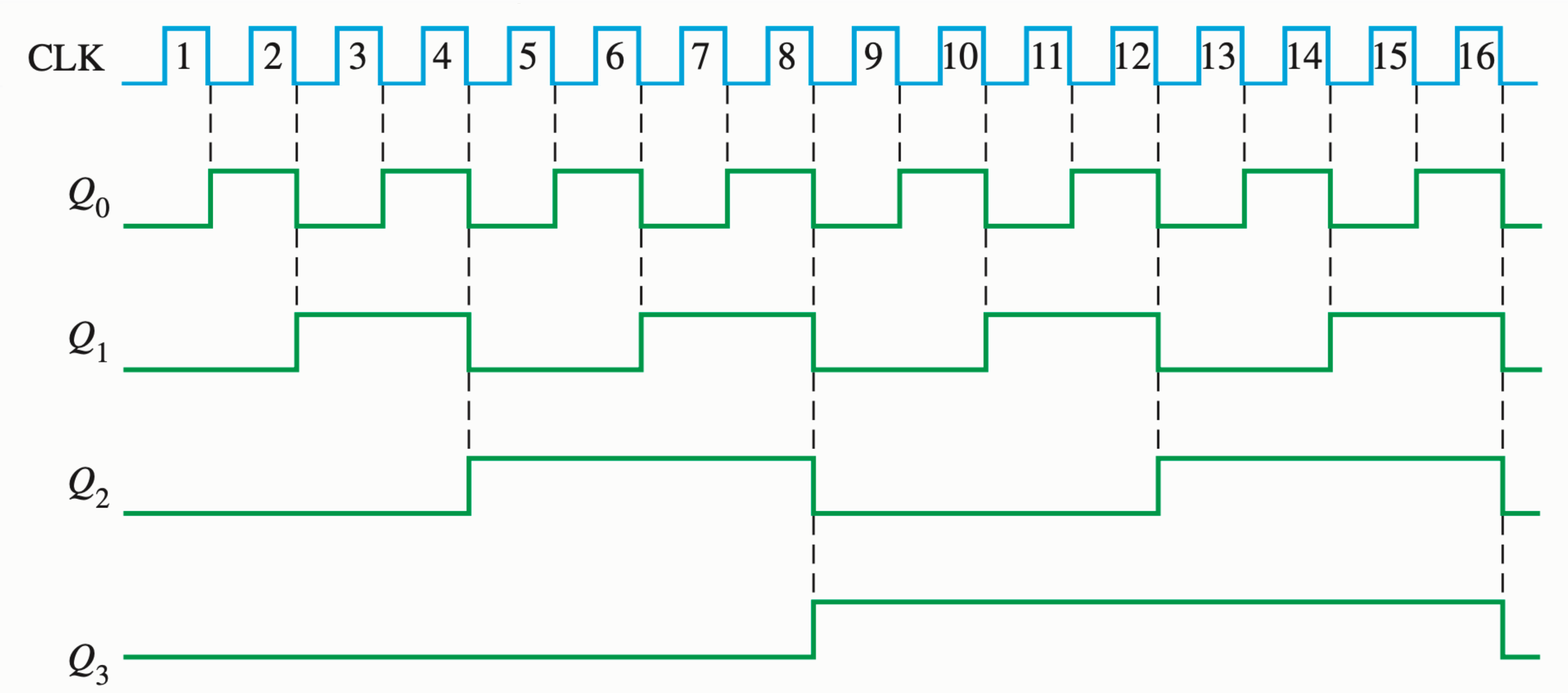
4-BIT BINARY RIPPLE COUNTER

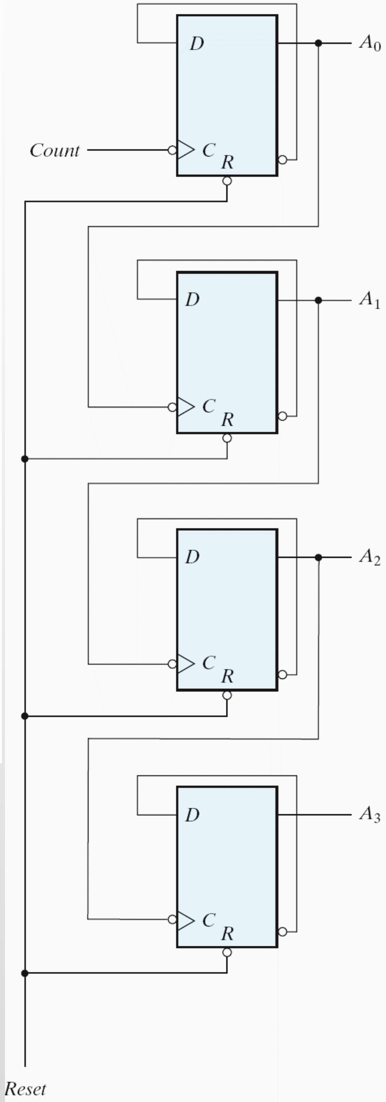
Binary Count Sequence

A_3	A_2	A_1	A_0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0

- It consists of a series connection of complementing flip-flops. The output of each flip-flop is connected to the *C* input of the next higher order flip-flops. It will count from 0 to 15 and then back to 0.





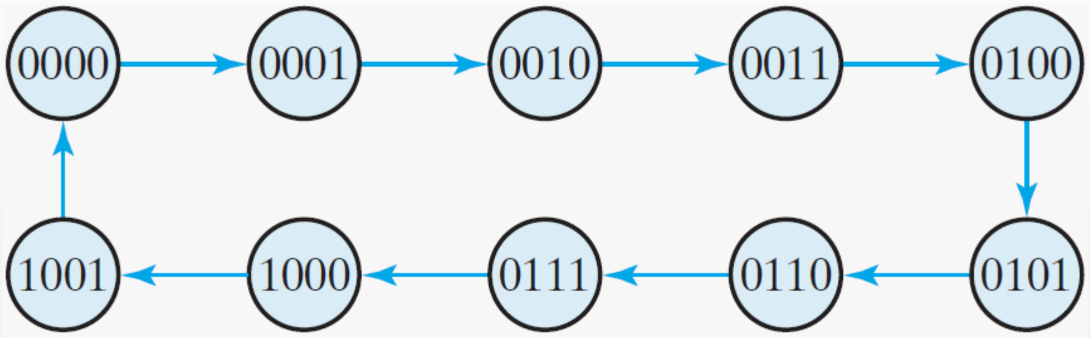


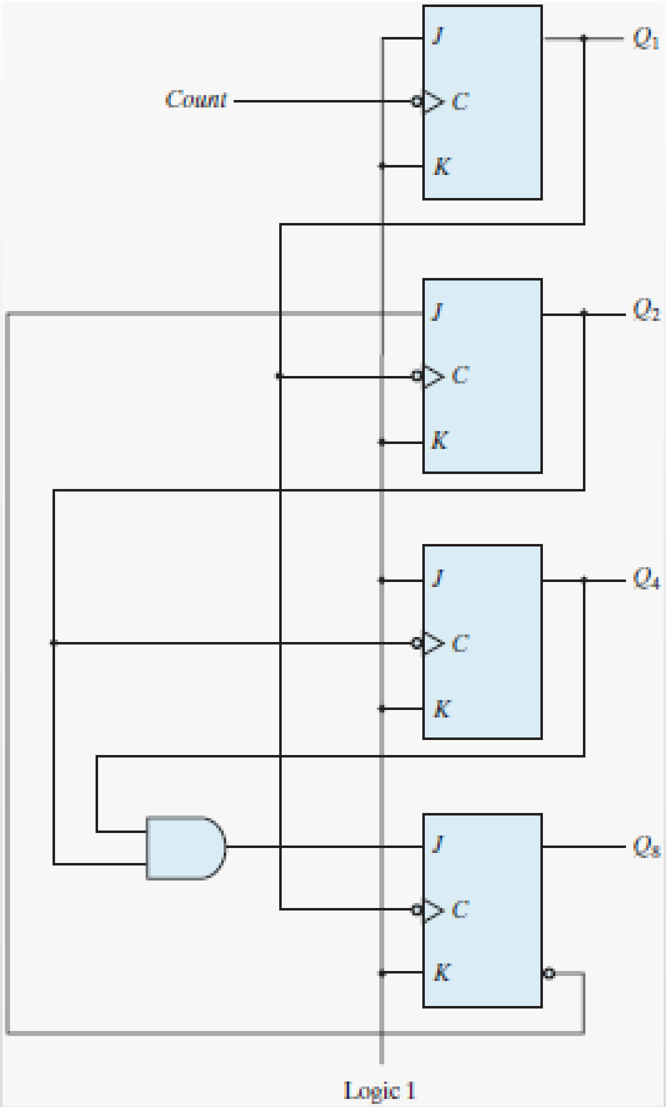
ACTIVITY

Draw the timing diagram of the 4-bit binary counter.

BINARY COUNTDOWN COUNTER

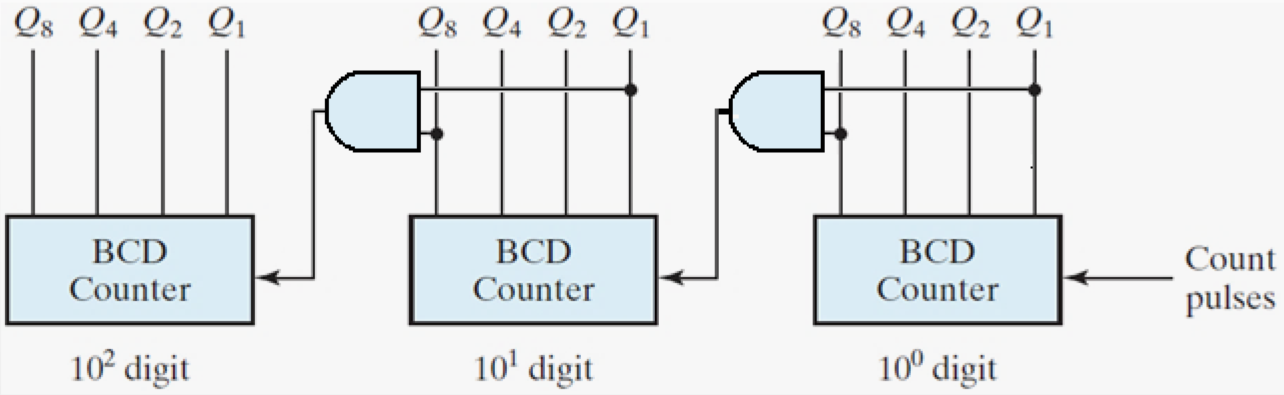
- A binary counter with a reverse count is called a binary countdown counter.
  - The binary count is decremented by 1 with every input count pulse.
  - The count of a 4-bit countdown counter starts from binary 15 and continues to binary counts 14, 13, 12, . . . , 0 and then back to 15.
- The diagram of a binary countdown counter looks the same as the binary ripple counter, provided that all flip-flops trigger on the positive edge of the clock (the bubble in the C inputs must be absent).

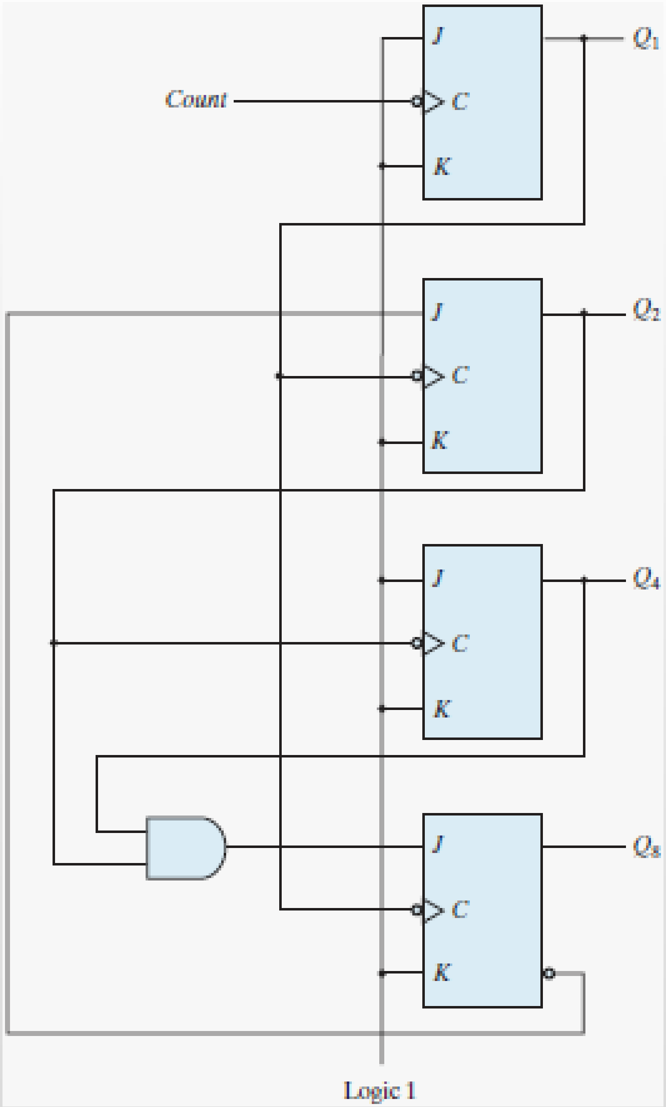




- Also known as a decade counter.
  - The final count is 9 before it goes back to 0.
- Remember that when the C input goes from 1 to 0, the flip-flop is set if J = 1, is cleared if K = 1, is complemented if J = K = 1, and is left unchanged if J = K = 0.

4-BIT BCD RIPPLE COUNTER





- When Q_8 in one decade goes from 1 to 0, it triggers the count for the next higher order decade while its own decade goes from 9 to 0.
- Multiple decade counters can be constructed by connecting BCD counters in cascade, one for each decade.

THREE-DECADE BCD RIPPLE COUNTER